

The Correlation of Omega-3 Supplementation with CRP As Marker Inflammation in Autoimmune Diseases

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ABSTRACT

Introduction: Omega-3 fatty acids are a type of polyunsaturated fatty acids (PUFAs) are regarded as anti-inflammatory lipids. The occurrence of diseases with an inflammatory aspect, including acute myocardial infarction, diabetes mellitus, autoimmune diseases such as multiple sclerosis (MS), rheumatoid arthritis (RA), psoriasis vulgaris, type 1 diabetes mellitus (T1DM), inflammatory bowel diseases (IBD) and systemic lupus erythematosus (SLE). Omega 3 can reduce inflammatory factor such as CRP (C- reactive protein) or hsCRP (High sensitive c-reactive protein).

Objective: This systematic review aimed to investigate the beneficial effects of omega 3 fatty acid supplements which reduce inflammatory markers such as CRP in autoimmune populations.

Materials and Methods: This systematic review was reported based on criteria from Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). A literature search was conducted with multiple electronic databases, such as PubMed, ScienceDirect, Google Scholar. The keywords used in electronic databases were described using Boolean operators. All the studies from these databases were stored in the authors' library in Rayyan.ai.

Results: Six studies were included in the qualitative synthesis. From six studies, 50% show omega-3 has a beneficial to reduce marker of inflammation such as CRP. But in the other case, 50% from the studies prove that omega-3 doesn't have effect to reduce CRP serum levels in patient with autoimmune.

Conclusion: This systematic review reveals that omega-3 supplementation for reducing CRP as marker inflammation in autoimmune disease has a potential. At the same time, this supplementation is associated with autoimmune recovery shown by the improvement of the CRP serum levels

KEYWORDS: C-reactive protein, Autoimmune, Omega-3 fatty acid, high-senstive CRP

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INTRODUCTION

Omega-3 fatty acids are a type of polyunsaturated fatty acids (PUFAs) that contain at least one double bond located between the third and fourth carbon from the omega end. At present, the three omega-3 polyunsaturated fatty acids (PUFAs) most pertinent to clinical practice are α -linolenic acid (ALA), eicosapentaenoic acid (EPA), and docosahexaenoic acid (DHA). These fatty acid-containing oils which derived from plants and are found in beans, seeds, nuts, fish, and green leafy vegetables.^[1]

There are two main types of PUFAs: Omega-3 and Omega-6. The main difference between them lies in the location of their double bonds on the carbon chain, found at either position 3 or 6. The three primary Omega-3 PUFAs are α -linolenic acid (ALA), eicosapentaenoic acid (EPA), and docosahexaenoic acid (DHA).^[2] Omega-3 fatty acids are regarded as anti-inflammatory lipids due to epidemiological research on Greenland Eskimos, whose diet is high in polyunsaturated fatty acids sourced from fish. The occurrence of diseases with an inflammatory aspect, including acute

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myocardial infarction, diabetes mellitus, autoimmune diseases such as multiple sclerosis (MS), rheumatoid arthritis (RA), psoriasis vulgaris, type 1 diabetes mellitus (T1DM), inflammatory bowel diseases (IBD) and systemic lupus erythematosus (SLE), asthma, and thyrotoxicosis, was reduced among Eskimos in comparison to populations in Western countries.^[3]

Metabolites derived from Omega-3 can modulate the innate immune response of B and T lymphocytes along with their cytokines, reducing the secretion of interleukin (IL)-2, tumor necrosis factor alpha (TNF- α), and interferon-gamma (IFN- γ) from CD8 positive T cells, managing the differentiation of CD4+ T-cells into T-helper 1 (Th1) and Th17, and enhancing IgG production by B cells. Omega 3 can reduce inflammatory factor such as CRP (C- reactive protein) or hsCRP (High sensitive c-reactive protein).^[3] Dietary long-chain Omega-3 fatty acids from marine sources reduce systemic inflammation and improve both clinical symptoms and laboratory results in certain autoimmune disorders. There is still insufficient evidence regarding the potential of Omega-3 fatty acids to reduce the risk of autoimmune diseases, as indicated in the VITAL trial.^[4]

A sub-study of VITAL study^[5] participants evaluated the changes in systemic inflammation biomarkers, such as IL-6, TNF receptor 2, and high sensitivity C reactive protein, and found no indications of reductions from Omega-3 alone without vitamin D during the initial 2 years. Alternative research in animals and in vitro suggests that higher dietary consumption of eicosapentaenoic acid and docosahexaenoic acid reduces the synthesis of C reactive protein (CRP) and inflammatory cytokines such as TNF- α , IL-1 β , and IL-6, in addition to diminishing T cell proliferation and activation while serving as a precursor for specialized pro-resolving lipid mediators, including resolvins, protectins, and maresins, which aid in the resolution of inflammation.

Although animal studies show encouraging results, the findings of interventional studies in humans with SLE have not been clear. Several studies have indicated a positive clinical impact of dietary n-3 PUFA supplementation in individuals with SLE.^[6] Conversely, other research has indicated that n-3 PUFA supplementation does not affect disease activity, shows no decrease in inflammatory markers in SLE.^[7]

In spite of the wide-range research on the implementation of omega 3 in autoimmune patients, a systematic review that specifically investigates omega 3 to lowering CRP as marker inflammation in autoimmune population has a gap result. Consequently, this study is aimed to investigate the beneficial effects of omega 3 fatty acid supplements which reduce inflammatory markers such as CRP in autoimmune populations.

MATERIALS AND METHODS

Study methodology

We adhered to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta Analyses) guidelines.

Eligibility criteria

The following criteria were considered for studies' eligibility: type of study, samples, outcome measure. Type of studies: Original research articles from 2015- March 2025. Narrative review, systematic review, meta-analysis, non-comparative research, in vivo and vitro only studies, technical reports, editor response, scientific poster, study protocols, and conference abstracts were excluded. Articles without full-text availability, non-English, and irrelevant topics were also excluded. Samples of autoimmune patients such as RA, MS, SLE, T1DM, Psoriasis vulgrais, IBD were included in this study.

Outcome measure

The primary outcomes were reduction of serum levels of marker inflammation C-reactive protein after diet or consuming omega 3 fatty acids in autoimmune population with or without therapy.

Data sources and search

A literature search process was carried out with multiple electronic databases, such as PubMed, ScienceDirect, Google Scholar, and Scopus. The search was conducted from the inception of the database until March 2025. The keywords used in electronic databases were described using Boolean operators. All the studies from these databases were stored in the authors' library in Rayyan.ai.

Studies Selection

After duplicates removal, retrieved articles were screened based on their titles and abstracts by second independent reviewers (IKSA, AAARD). Potentially eligible full-text articles were thoroughly assessed using the eligibility criteria described above. Any emerging discrepancies were resolved by consensus among the review team. The study selection process was recorded in the PRISMA flow chart.

Data Extraction and Analysis

Selected studies were extracted with Microsoft Excel 2016 (Microsoft Corporation, USA) and Rayyan.ai. The following data were recorded: first author, year, country, study design, sample size, autoimmune type, aim, Omega 3 type, dose, assessment period, and outcome.

RESULT

Study selection

Our search yielded 354 articles from 3 databases as described in **Table 1** below. From hand searching we found 6 articles that matched with our topics. Twenty five duplicates from those databases were removed. Then, the authors read the titles and abstracts of the remaining three hundred and twentynine articles for preliminary screening. The authors excluded the articles that did not fulfill our eligibility criteria. Full-texts were retrieved for 124 articles,

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and 67 studies were excluded because those articles does not clearly explain the effect of omega 3 in autoimmune patients. Full-texts were retrieved again for 57 studies, and 27 studies were excluded because the studies don't explain clearly the inflammation marker of CRP. Full-texts 6 articles were

included to qualitative systematic reviews (sixteen studies from database selection and four studies from hand searching process). Our study selection process was presented in the PRISMA diagram on **Figure 1**.

Table 1. Database Searching Process Result

Database	Keyword	Articles
PubMed	(((((Omega 3) OR (fatty acid omega)) OR (PUFA)) OR (fish oil)) AND (CRP)) OR (C-reactive protein)) OR (high sensitive CRP)) AND (autoimmune)	29
Sciendirect	("Omega 3" OR "PUFA" OR "Fish oil") AND ("CRP" OR "C-reactive protein" OR "hs-CRP" OR "high sensitive CRP") AND ("Autoimmune")	321
Google Scholar	("Omega 3" OR "PUFA" OR "Fish oil") AND ("CRP" OR "C-reactive protein" OR "hs-CRP" OR "high sensitive CRP") AND ("Autoimmune") OR ("Rheumatoid arthritis" OR "Systemic Lupus Erytematosus" OR "multiple Sceloris" OR "Sickle cell")	4

Study characteristics and results of individual studies

The full details of each study were displayed in Table 2. The subject of these studies was people with autoimmune disease such as Rheumatoid arthritis (RA), Type 1 Diabetic Mellitus (T1DM), and Systemic Lupus Erythematosus (SLE) which are diet with omega-3. We focused on CRP and hs-CRP as markers of inflammation to indicate the effectiveness of omega-3 to lower serum levels of CRP. Out of the 6 studies included in the qualitative synthesis, 1 study of the sample was pediatric.

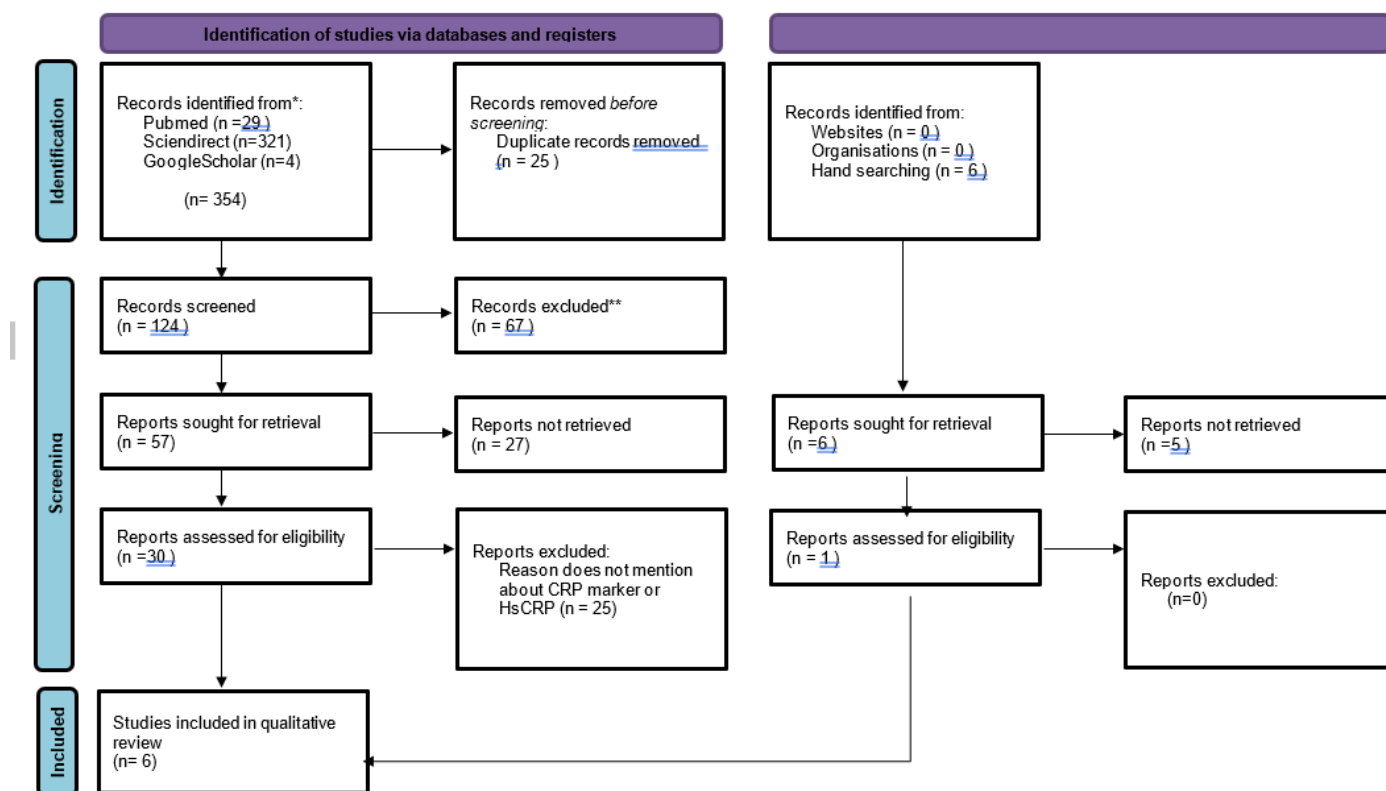


Figure 1. PRISMA 2020 Flow Diagram

DISCUSSION

CRP as a Marker Inflammation in Autoimmune Disease

C-reactive protein is a homopentameric protein involved in the acute-phase inflammatory response, a highly conserved plasma protein identified in 1930 by Tillet and Francis during their study of the sera from patients with the acute phase of *Pneumococcus* infection; it was named due to its reaction with the capsular (C)-polysaccharide of *Pneumococcus*.^[8] Increased expression of C-reactive protein is seen in inflammatory diseases such as rheumatoid arthritis, infection, and some cardiovascular disorders. CRP is an acute-phase protein, and during inflammatory diseases, its plasma concentration varied by at least 25%.

Rheumatoid arthritis is known to be linked with chronic inflammation and indicators of systemic inflammation, including C-reactive protein (CRP) and erythrocyte sedimentation rate (ESR).^[9] These laboratory results, including erythrocyte sedimentation rate (ESR), C-reactive protein (CRP), and rheumatoid factor (RF), significantly influence the prognosis of RA. Traditionally, the erythrocyte sedimentation rate (ESR) has been the most commonly utilized marker of inflammation in RA; however, ESR levels react slowly to inflammatory triggers and, consequently, to fluctuations in disease activity. The function of CRP in rheumatoid arthritis: CRP serves as more than just an indicator of inflammation or infection; it acts as an immunological modulator. CRP is an acute phase protein produced by hepatocytes and released into the bloodstream as pentameric CRP (pCRP) in response to specific proinflammatory cytokines, particularly IL-6.^[10]

In patient with SLE, CRP more as a marker for serious infection. It is important to examine the function of CRP in SLE. CRP is directly influenced by IL-6 and IL-6 concentration are elevated in active SLE. In the other context, CRP in SLE is frequently not completely normal. The higher CRP are observed in individuals with active serositis, arthritis, or myositis. In the other cases, CRP level typically stay under 60mg/L or 6 mg/dl in people with active SLE. The above of level CRP in indicated in serious infection.^{[11][12][13]}

Diet Source of Omega

Omega it can be taken from nuts and fish, especially small oily fish such as anchovies, sardines, or mackerel. ALA is present in plant sources, nuts, and seeds, whereas EPA and

DHA are the primary elements of fish oil. Omega-6, such as LA and AA, are present in refined vegetable oils, meats, poultry, eggs, and dairy products. Omega-3 possesses anti-inflammatory characteristics, whereas Omega-6 exhibits pro-inflammatory effects; however, achieving a balance between the two is essential for nutritious health. The Western diet has significantly altered over the past 30 years, in numerous harmful ways, including a change towards a greater Omega-6/Omega-3 ratio. Indeed, the diets in industrialized nations are currently primarily characterized by Omega-6 PUFAs and have minimal quantities of Omega-3 PUFAs, with the Omega-6/Omega-3 ratio soaring to 20-30, a condition associated with chronic inflammation. Eating less seafood and fewer whole foods reduces Omega-3 consumption. The reduced intake of seafood, especially fresh options high in Omega-3 PUFA such as salmon, trout, crabs, oysters, and mussels, may be partly linked to the decreased affordability of this diet.^[14] From our study, most of them use omega 3 as intervention for populations. There Is only one study from Cecilia et al which uses omega-3 and omega-6 for intervention in study.

In the study Yuanqing et al^[15] use Hard-shelled mussels for diet. Hard-shelled mussels were gathered from China produce in November 2011. The product forms dried powder after the process of vacuum freeze. HMLE supplementation notably enhanced paw swelling, arthritic index, and body weight gain. Additional analysis revealed that HMLE decreased the expression of pro-inflammatory factors such as PGE2 and IL-1 β . IL-6 and TNF- α , along with elevated levels of anti-inflammatory cytokines like IL-4 and IL-10, which suggested that the notable effectiveness could be linked to managing the equilibrium between pro-inflammatory and anti-inflammatory elements.^[16] The other study of Helen et al^[17] uses a blue mussel diet. Blue mussel diet related with fish and the composition are eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA). Furthermore, blue mussel abundant in nutrients like zinc, selenium, riboflavin, and carotenoids. Fish and shellfish are frequently seen as a uniform food category, despite considerable differences in their nutrient content. A recent investigation of lipid extracts from the hard-shelled mussel (*Mytilus coruscus*) in individuals with RA indicated decreased disease activity and enhancements in certain cytokines.^[18] Apart from EPA and DHA, new bioactive fatty acids have been discovered in blue mussel and green lipped mussel (*Perna canaliculus*).

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Table 2. Qualitative summary of literature study about omega-3 correlation with CRP

Study	Helen et al ^[17]	Mariane et al ^[7]	Cecilia et al ^[19]	Elham et al ^[20]
Country	Sweden	Brazil	Sweden	Iran
Year	2018	2017	2017	2015
Study Design	Randomized control trial	Randomized Control Trial	Case control study	Randomized Control Trial
Sample	n= 39 (Blue mussel diet= 20, control diet=19)	n= 66 (Omega-3 Group= 22, Control group= 27, Did not complete the trial= 17)	n= 591	n=60 (Omega-3 Group=30, control group=30)
Autoimmune type	RA	SLE	RA	RA
Aim	To identify a beneficial from diet in blue mussels to reduce disease activity in patient with RA.	To identify the correlation of omega-3 on marker inflammatory in SLE patient.	To investigate the effectiveness of dietary PUFA in pain pattern patient with RA.	To evaluation effect of omega-3 on disease activity of RA.
Diet	Blue mussel	Omega-3	Omega-3 and Omega-6	Omega-3
Dose	75-g portion of blue mussels that contain: EPA (fatty acid 20:5) 0.18 g and DHA (fatty acid 22:6) 0.16 g	540 mg of EPA and 100 mg of DHA	omega-3 FA: 1st tertile: ≤0.48 g/day 2nd tertile: 0.490.77 g/day 3rd tertile: >0.78 g/day Omega-6 FA 1st tertile: ≤6.00 g/day 2nd tertile: 6.018.19 g/day 3rd tertile: >8.20 g/day	1.8 gram EPA and 2.1 gram DHA
Assesment periode	2 × 11 weeks	3 months	3 months	3 months
Outcome	Control: Before 2.0 (0.0, 3.0) After 2.0 (0.0, 3.0) Δ a 0.0 (−1.0, 1.0) Blue Mussel diet: Before 2.0 (1.0, 5.0) After 1.0 (1.0, 3.0) Δ a −1.0 (−1.0, 1.0) P= 0.106	CRP levels Omega-3 T0 5.0 (4.9–8.1); T1 4.9 (4.9–7.2) p=0.230 CRP levels control T0 6.4 (4.9–11.6); T1 5.0 (4.9–11.6) p=0.009	CRP Levels omega-3 FA/ fish oil (1.10 (0.73–1.65) p=0.665)	CRP Omega-3 baseline +2, end study +0 to +1; CRP control baseline +2, end study +2 to +3

Study	H. Peter et al ^[21]	Yuanqing et al ^[15]
Country	USA	China
Year	2015	2015
Study Design	Randomized control trial	Randomized Control Trial
Sample	n= 37 Group A control (n=16) and treatment (n=21)	n= 50 (Omega 3 group=25; control group=25)
Autoimmune type	Type 1 Diabetic melitus	RA
Aim	To determine the effect of DHA supplementation, in infants with a high genetic risk for developing T1D	To investigated HMLE with improvement of clinical condition from RA patients.
Diet	Omega-3	Hard-shelled mussels
Dose	Group A =DHA (800 mg/day) or corn/soy oil (800 mg/day).	Group A= four HMLE capsules (200 mg HMLE and 200 mg corn oil per capsule) once daily after meals for

	Formula-fed infants= formula + 10.2 mg DHA/ounce (treatment) or 3.4 mg DHA/ounce (control) At age 12 months, all infants were treated with two 200-mg capsules per day of either DHA or corn/soy oil (per earlier randomization) until age 36 months	first 2 months and 2 HMLE capsules once daily after meals for subsequent 4 months Group B= orally placebo capsules (400 mg corn oil per capsule, identical in appearance to HMLE capsules) according to the same schedule.
Assesment periode	36 months	3 months
Outcome	Group A hsCRP level at 12 months for the breast-fed treatment infants was 0.012 [0.003, 0.048]. Group B The mean hsCRP level for all formula-fed infants (control and treatment) at 12 months was 0.086 (P=0.007)	CRP HMLE Group : Men= (16.0 ± 4.4); women= (13.7 ± 1.5) CRP Controp Group: Men= (11.9 ± 3.5); Women=(17.8 ± 2.3) P= 0.490

CORRELATION SUPPLEMENTATION DOSAGE IN CRP LEVELS

The supplementation dosage of Omega-3 that we used in our study was variation. The omega-3 types that were used were DHA and EPA in studies from Mariane et al and Elham et al. Another study from cecilia et al does not mention the type of Omega-3. And in study from Peter et al, only used DHA. The dosage of DHA and EPA varied immensely in the 5 included studies. The dosage varies from 3.4 mg to 2100 mg DHA per day, and from 100 mg to 1800 mg EPA per day. Supplementation could either contain only DHA or DHA in combination with EPA. And the only one of study using HMLE with 200 mg dosage.

The studies from Mariane et al in which participants received supplementation of 540 mg of EPA and 100 mg of DHA did not show a positive effect. The median (IQR) in CRP levels variation between the two groups is represented in **Fig. 3**, showing a decrease in the omega-3 group while there was an increase in the control group (p = 0.008). The result of this study was related to Cecilia et al that found no association between use of omega3 FA/fish oil supplementation and refractory pain or inflammatory parameters (CRP). This study shows an impact of omega-3 on CRP levels, as there was a notable difference in serum levels among groups: the omega-3 group experienced a decrease, whereas the control group had an increase. Research conducted on healthy individuals and on patients with chronic inflammatory conditions such as diabetes, coronary heart disease, ulcerative colitis, dyslipidemia, and rheumatoid arthritis has demonstrated that the intake of omega-3 fatty acids is negatively correlated.

Omega-3 PUFAs may lessen inflammation via various mechanisms. They reduce the synthesis of eicosanoid mediators derived from arachidonic acid, known for their pro-inflammatory effects. Additionally, the enzymatic oxidation

of EPA and DHA generates resolvins and proteins, which assist in resolving inflammation. In vitro and in vivo research indicated that this mechanism might require Omega-3 fatty acid levels as high as 3 g.^[22] There were some studies indicating that omega-3 fatty acid supplementation reduces inflammatory markers like CRP, IL-6, and TNF levels. This idea was open because omega-3 fatty acid-containing cod liver oil supplements can serve as a sparing agent for non-steroidal anti-inflammatory medications in patients with rheumatoid arthritis.^[19] While the result about CRP when using omega-3 FA was 1.10 (0.73–1.65) p-value =0.665. The result proved that it is non significance.

In the other study of Elham et al with a higher supplementation dosage (i.e., 1.8 gram EPA and 2.1 gram DHA per day) a significant effect was shown. CRP is primarily utilized to monitor disease activity in the acute phase for individuals with rheumatoid arthritis. Some studies have demonstrated that Omega-3 intake plays a significant role in reducing inflammatory markers.^[23] The study of Peter et al shows the same result with Elham et al. In their experiences the level of hsCRP was significantly lower (P=0.007) at age 12 months in the present study in breast-fed infants whose mothers received DHA compared to all formula-fed infants (with or without DHA). High sensitive C reactive protein (hsCRP) is recognized as a major factor and a low-grade inflammatory marker involved in the advancement of vaso-occlusive crisis (VOC) in children. hsCRP is also viewed as a critical connection between atherosclerosis, cardiovascular disease (CVD), and insulin resistance.^{[24][25]} Study from Pilar Perez et al, 2020 [24] shows the elevated hsCRP levels found in children with type 1 diabetes, when compared to a control group with comparable BMI, indicate a baseline inflammatory condition that may heighten cardiovascular risk.

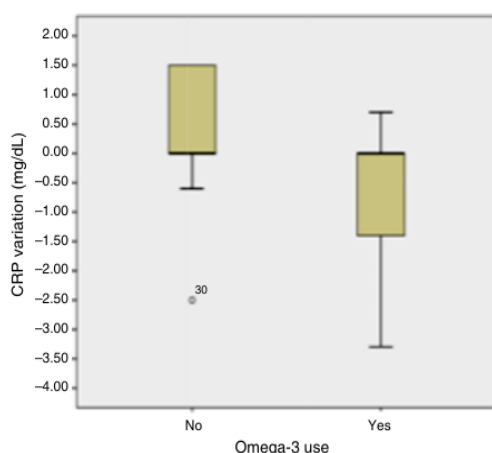


Figure 2. The correlation between omega-3 use with CRP variation

Moreover, note that in one study from Yuanqing et al, there are no relationship between HMLE and placebo groups for CRP serum levels. CRP HMLE Group: Men= (16.0 ± 4.4); women= (13.7 ± 1.5); CRP Control Group: Men= (11.9 ± 3.5); Women= (17.8 ± 2.3) P value = 0.490. The result proved that the use of HMLE was not significant for improving CRP serum levels. In contrast, results from Helen et al show the impact of blue mussel diet for reducing CRP serum level although not significant. The result was different from Fu et al, 2015 [18] where, lipid extract which from hard-shelled mussels consumes in patient with RA. The evaluating from that show there is no effect were seen on either ESR or CRP, but DAS28 (calculated without VAS GH), tumor necrosis factor α , interleukin 10 and prostaglandin E2 were improved after six months.

STRENGTH AND LIMITATION

This review had a number of limitations. From our review, we have to conclude that there is large variability in the studies. The different count of sample it can be bias from this review. The studies differed on various factors: the study population (i.e., age, type of autoimmune disease); composition of supplements (i.e., EPA/DHA ratio, but also inclusion of other fatty acids or nutrients); and duration of supplementation. Despite the limitations of the present study, our result suggests that it is the first review in which the relationship between omega-3 and CRP serum levels in children and adolescents which has autoimmune disease. The intervention was carefully performed, and the outcome measurements are clinically relevant.

CONCLUSION

This systematic review reveals that omega-3 supplementation for reducing CRP as marker inflammation in autoimmune disease shows 50% potential. At the same time, this supplementation is associated with autoimmune recovery shown by the improvement of the CRP serum levels. The

future systematic review prefers to have the same population in type of disease. So the review will be a comprehensive and more sensitive result for the CRP serums level. And maybe in the next review, we can add more marker inflammation in autoimmune diseases such as cytokines pro-inflammation.

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CONFLICT OF INTEREST

No potential conflict of interest was reported by the authors

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